SF1811 Optimization
A note on optimality conditions

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First-order necessary optimality conditions are given for equality-constrained problems and inequality-constrained problems separately in the textbook. In this note, they are given jointly, which is straightforward for linear constraints. The relationship between linear constraints and linearized nonliner constraints is given. For completeness, regularity conditions for the nonlinearly constrained case are given. This is for the student who is interested. In addition, a few comments related to homework assignment 3 are given.

A. Linearly constrained problems

Consider the problem

$$(P_{lin}) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & b_i - a_i^T x = 0, \ i \in \mathcal{E}, \\ & b_i - a_i^T x \leq 0, \ i \in \mathcal{I}, \\ & x \in I\!R^n, \end{array}$$

where f is continuously differentiable and $\mathcal{E}$ and $\mathcal{I}$ are nonintersecting index sets. This means that $\mathcal{E} \cup \mathcal{I} = \{1, \ldots, m\}$ and $\mathcal{E} \cap \mathcal{I} = \emptyset$. Problem (P_{lin}) includes equality-constrained problems, $\mathcal{I} = \emptyset$, inequality-constrained problems, $\mathcal{E} = \emptyset$, and unconstrained problems, $\mathcal{E} = \mathcal{I} = \emptyset$, as special cases.

The first-order necessary optimality conditions are based on the statement that there cannot exist a feasible descent direction at a local minimizer. This is summarized in the following lemma.

Lemma A.1. *Assume that $\hat{x}$ is a local minimizer to (P_{lin}). Then, there cannot exist a d such that*

$$\begin{aligned} \nabla f(\hat{x})d &< 0, \\ a_i^T d &= 0, \quad i \in \mathcal{E}, \\ a_i^T d &\geq 0, \quad i \in \mathcal{I}_a(\hat{x}), \end{aligned}$$

where $\mathcal{I}_a(\hat{x}) = \{i \in \mathcal{I} : b_i - a_i^T \hat{x} = 0\}$.

Proof. The proof is by contradiction. Assume such a d exists. Then, $\nabla f(\hat{x})d < 0$ implies $f(\hat{x} + td) < f(\hat{x})$ for t sufficiently small and positive. We now show that $\hat{x} + td$ is feasible to (P_{lin}) for t sufficiently small and positive. For $i \in \mathcal{E}$ and $t > 0$, we have

$$b_i - a_i^T(\hat{x} + td) = b_i - a_i^T \hat{x} - t a_i^T d = 0,$$

as $b_i - a_i^T \hat{x} = 0$ and $a_i^T d = 0$. Similary, for $i \in \mathcal{I}_a(\hat{x})$, we have

$$b_i - a_i^T(\hat{x} + td) = b_i - a_i^T \hat{x} - t a_i^T d \leq 0,$$

as $b_i - a_i^T \hat{x} = 0$, $a_i^T d \geq 0$ and $t > 0$. Finally, for $i \in \mathcal{I} \backslash \mathcal{I}_a(\hat{x})$, we have

$$b_i - a_i^T(\hat{x} + td) = b_i - a_i^T \hat{x} - t a_i^T d < 0,$$

for $t \geq 0$ if $a_i^T d \geq 0$, and for $0 \leq t < -(b_i - a_i^T x)/(-a_i^T d)$ if $a_i^T d < 0$. Note that this upper bound is strictly positive, as $b_i - a_i^T \hat{x} < 0$ for $i \in \mathcal{I} \backslash \mathcal{I}_a(\hat{x})$. We conclude that $\hat{x} + td$ is feasible to (P_{lin}) for

$$0 \leq t < \min_{i \in \mathcal{I} \backslash \mathcal{I}_a(\hat{x}) : a_i^T d < 0} \frac{-(b_i - a_i^T x)}{-a_i^T d},$$

where the upper bound is infinite if there is no $i \in \mathcal{I} \backslash \mathcal{I}_a(\hat{x})$ such that $a_i^T d < 0$.
We conclude that such a d would be a feasible descent direction, contradicting $\hat{x}$ being a local minimizer to (P_{lin}). Consequently, if $\hat{x}$ is a local minimizer to (P_{lin}), no such d can exist. ∎

Rather than verifying that there exists no d with the properties of Lemma A.1, we can use strong duality for linear programming, alternatively Farkas lemma, to give an equivalent condition involving existence of Lagrange multipliers, as stated in the following lemma.

Lemma A.2. *Exactly one of the following two systems has a solution:*

1. $\nabla f(\hat{x})d < 0$, $a_i^T d = 0$, $i \in \mathcal{E}$, $a_i^T d \geq 0$, $i \in \mathcal{I}_a(\hat{x})$,
2. $\sum_{i\in\mathcal{E}} a_i u_i + \sum_{i\in\mathcal{I}_a(\hat{x})} a_i y_i = \nabla f(\hat{x})^T$, $y_i \geq 0$, $i \in \mathcal{I}_a(\hat{x})$.

Proof. Consider the following primal-dual pair of linear programs

$$
(PLP)\quad \begin{array}{ll} \text{minimize} & \nabla f(\hat{x})d \\ \text{subject to} & a_i^T d = 0,\ i \in \mathcal{E}, \\ & a_i^T d \geq 0,\ i \in \mathcal{I}_a(\hat{x}), \end{array}
\qquad
(DLP)\quad \begin{array}{ll} \text{maximize} & \sum_{i\in\mathcal{E}} 0u_i + \sum_{i\in\mathcal{I}_a(\hat{x})} 0y_i \\ \text{subject to} & \sum_{i\in\mathcal{E}} a_i u_i + \sum_{i\in\mathcal{I}_a(\hat{x})} a_i y_i = \nabla f(\hat{x})^T, \\ & y_i \geq 0,\ i \in \mathcal{I}_a(\hat{x}). \end{array}
$$

It can be verified by Lagrangian relaxation that these are primal-dual pairs, or they can be taken from the general formula on the formula sheet.
Consider (PLP). A feasible point is given by $d = 0$. Therefore, the optimal value of (PLP) is less than equal to zero. In addition, the optimal value is zero if and only if there is no d that satisfies 1. If there is a d that satisfies 1, the optimal value of (PLP) is $-\infty$. This is a consequence of a d that satisfies 1 implies that td satisfies 1 for $t > 0$. Then, $\nabla f(\hat{x})d < 0$ implies

$$\lim_{t\to\infty} \nabla f(\hat{x})(td) = \nabla f(\hat{x})d \lim_{t\to\infty} t = -\infty.$$

We have proved strong duality for linear programming in the course. This means that if the optimal value of (PLP) is zero, i.e., if 1 has no solution, the optimal value of (DLP) is zero, i.e., 2 has a solution. If the optimal value of (PLP) is $-\infty$, i.e., if 1 has a solution, then (DLP) is infeasible, which means that 2 has no solution. ∎

A combination of these two lemmas then give the first-order necessary optimality conditions, as stated in the following theorem.

Theorem A.1. *Assume that $\hat{x}$ is a local minimizer to (P_{lin}). Then there exist $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{I}$, such that $\hat{x}$, $\hat{u}$ and $\hat{y}$ satisfy*

1. $\nabla f(\hat{x}) - \sum_{i\in\mathcal{E}} u_i a_i^T - \sum_{i\in\mathcal{I}} y_i a_i^T = 0$,
2. $b_i - a_i^T\hat{x} = 0$, $i \in \mathcal{E}$, $\quad b_i - a_i^T\hat{x} \leq 0$, $i \in \mathcal{I}$,
3. $\hat{y}_i \geq 0$, $i \in \mathcal{I}$,
4. $\hat{y}_i(b_i - a_i^T\hat{x}) = 0$, $i \in \mathcal{I}$.

Proof. Assume that $\hat{x}$ is a local minimizer to (PLP). Then, Lemma A.1 and Lemma A.2 show the existence of $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{E}$, such that

$$\nabla f(\hat{x}) - \sum_{i\in\mathcal{E}} \hat{u}_i a_i^T - \sum_{i\in\mathcal{I}_a(\hat{x})} \hat{y}_i a_i^T = 0, \quad \hat{y}_i \geq 0,\ i \in \mathcal{I}_a(\hat{x}).$$

By noting that $\hat{x}$ is feasible to (P_{lin}), and by defining $\hat{y}_i = 0$, $i \in \mathcal{I}\backslash\mathcal{I}_a(\hat{x})$, 1–4 are satisfied. ∎

These are the conditions you need for homework assignment 3.

B. Nonlinearly constrained problems

Consider the problem

$$(P)\qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0,\ i \in \mathcal{E}, \\ & g_i(x) \leq 0,\ i \in \mathcal{I}, \\ & x \in I\!R^n, \end{array}$$

where f, g_i, $i \in \mathcal{I}$ and h_i, $i \in \mathcal{I}$, are continuously differentiable and $\mathcal{E}$ and $\mathcal{I}$ are nonintersecting index sets.

The first-order necessary optimality conditions may be viewed as necessary conditions for $\hat{x}$ to be a local minimizer to the problem where the constraint functions are replaced by their linearization at $\hat{x}$, i.e.,

$$(\hat{P}_{lin})\qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(\hat{x}) + \nabla h_i(\hat{x})(x - \hat{x}) = 0,\ i \in \mathcal{E}, \\ & g_i(\hat{x}) + \nabla g_i(\hat{x})(x - \hat{x}) \leq 0,\ i \in \mathcal{I}, \\ & x \in I\!R^n. \end{array}$$

To put $(\hat{P}_{lin})$ into the notation of the previous section, we may identify

$$b_i = h_i(\hat{x}) - \nabla h_i(\hat{x})\hat{x}, \quad a_i^T = -\nabla h_i(\hat{x}), \quad i \in \mathcal{E}, \tag{B.1a}$$

$$b_i = g_i(\hat{x}) - \nabla g_i(\hat{x})\hat{x}, \quad a_i^T = -\nabla g_i(\hat{x}), \quad i \in \mathcal{I}. \tag{B.1b}$$

For this linearly constrained problems, we may apply the analysis of the previous section. In particular, Theorem A.1 gives the following corollary.

Corollary B.1. *Assume that $\hat{x}$ is a local minimizer to $(\hat{P}_{lin})$. Then there exist $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{I}$, such that $\hat{x}$, $\hat{u}$ and $\hat{y}$ satisfy*

1. $\nabla f(\hat{x}) + \sum_{i\in\mathcal{E}} \hat{u}_i \nabla h_i(\hat{x}) + \sum_{i\in\mathcal{I}} \hat{y}_i \nabla g_i(\hat{x}) = 0$,
2. $h_i(\hat{x}) = 0$, $i \in \mathcal{E}$, $\quad g_i(\hat{x}) \leq 0$, $i \in \mathcal{I}$,
3. $\hat{y}_i \geq 0$, $i \in \mathcal{I}$,
4. $\hat{y}_i g_i(\hat{x}) = 0$, $i \in \mathcal{I}$.

If the linearization of the constraints at $\hat{x}$ describe the nonlinear constraints sufficiently well in a neighborhood of $\hat{x}$, the first-order necessary optimality conditions for $(\hat{P}_{lin})$ are also first-order necessary conditions for (P). If this is the case, we say that $\hat{x}$ is a regular point. Therefore, we have the following result.

Theorem B.1. *Assume that $\hat{x}$ is a local minimizer to (P) which is also a regular point. Then there exist $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{I}$, such that $\hat{x}$, $\hat{u}$ and $\hat{y}$ satisfy*

1. $\nabla f(\hat{x}) + \sum_{i\in\mathcal{E}} \hat{u}_i \nabla h_i(\hat{x}) + \sum_{i\in\mathcal{I}} \hat{y}_i \nabla g_i(\hat{x}) = 0$,
2. $h_i(\hat{x}) = 0$, $i \in \mathcal{E}$, $\quad g_i(\hat{x}) \leq 0$, $i \in \mathcal{I}$,
3. $\hat{y}_i \geq 0$, $i \in \mathcal{I}$,
4. $\hat{y}_i g_i(\hat{x}) = 0$, $i \in \mathcal{I}$.

Theorem B.1 states regularity as the condition needed to ensure that there exist $\hat{u}$ and $\hat{y}$ which together with $\hat{x}$ satisfy conditions 1–4 if $\hat{x}$ is a local minimizer. This is the result we want to use. However, in proving properties that ensure regularity, it may be easier to use the equivalent statement of the following corollary. This is a direct consequence of Lemma A.2.

Corollary B.2. *Assume that $\hat{x}$ is a feasible point to (P). In addition, assume that there exists a $\hat{d}$ such that*

$$\nabla f(\hat{x})\hat{d} < 0, \quad \nabla h_i(\hat{x})\hat{d} = 0,\ i \in \mathcal{E}, \quad \nabla g_i(\hat{x})\hat{d} \leq 0,\ i \in \mathcal{I}_a(\hat{x}),$$

Then, if $\hat{x}$ is a regular point, it cannot be a local minimizer to (P).

Proof. Lemma A.2 shows that for a given $\hat{x}$, which is feasible to (P), there always exist either $\hat{u}$ and $\hat{v}$ sastisfying the properties of Theorem B.1, or a $\hat{d}$ satisfying the properties of this corollary. Therefore, as regularity is defined as properties that make the properties of Theorem B.1 hold at a local minimizer, it is equivalent to properties that make the properties of this corollary not hold at a local minimizer. ∎

There are different conditions for ensuring that $\hat{x}$ is a regular point. They can be shown by verifying that a $\hat{d}$ satisfying the properties of Corollary B.2 implies that $\hat{x}$ is not a local minimizer. We give a few here:

1. *If the constraints are linear, i.e., $h_i(x) = b_i - a_i^T x$, $i \in \mathcal{E}$, and $g_i(x) = b_i - a_i^T x$, $i \in \mathcal{I}$, then every feasible point $\hat{x}$ is regular.*

This is the result of the previous section.

2. *If $\hat{x}$ is feasible, $\mathcal{I} = \emptyset$, and $\nabla h_i(\hat{x})$, $i \in \mathcal{E}$, are linearly independent, then $\hat{x}$ is a regular point.*

This is shown in the textbook.

3. *If $\hat{x}$ is feasible, $\mathcal{E} = \emptyset$, and there exists a d such that $\nabla g_i(\hat{x})d < 0$, $i \in \mathcal{I}$, then $\hat{x}$ is a regular point.*

This is shown in the textbook.

4. *If $\hat{x}$ is a feasible point such that $\nabla h_i(\hat{x})$, $i \in \mathcal{E}$ and $\nabla g_i(\hat{x})$, $i \in \mathcal{I}_a(\hat{x})$, are linearly independent, then $\hat{x}$ is a regular point.*

Proof. Consider the equality-constrained problem

$$(\hat{P}_=) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0,\ i \in \mathcal{E}, \\ & g_i(x) = 0,\ i \in \mathcal{I}_a(\hat{x}). \end{array}$$

If $\hat{x}$ is a local minimizer to (P), it is also a local minimizer to $(\hat{P}_=)$. Let d be such that $\nabla h_i(\hat{x})d = 0$, $i \in \mathcal{I}$, $\nabla g_i(\hat{x})d = 0$, $i \in \mathcal{I}_a(\hat{x})$. Then, Lemma C.1 shows the existence of an $\epsilon > 0$ and function $x(t)$ such that $x(0) = \hat{x}$, $\dot{x}(0) = d$, $h_i(x(t)) = 0$, $i \in \mathcal{I}$, and $g_i(x(t)) = 0$, $i \in \mathcal{I}_a(\hat{x})$, for $t \in [-\epsilon, \epsilon]$. Therefore, as $x(t)$ is feasible for $t \in [-\epsilon, \epsilon]$, we must have

$$0 = \frac{d}{dt} f(x(t))|_{t=0} = \nabla f(\hat{x})d.$$

Consequently, there must exist Lagrange multipliers corresponding to Theorem B.1, i.e.,

$$\nabla f(\hat{x}) + \sum_{i \in \mathcal{E}} \hat{u}_i \nabla h_i(\hat{x}) + \sum_{i \in \mathcal{I}_a(\hat{x})} \hat{y}_i \nabla g_i(\hat{x}) = 0,$$

but where the sign on $\hat{y}_i$, $i \in \mathcal{I}_a(\hat{x})$ has not been restricted. To show the sign, let d be such that $\nabla f(\hat{x})d < 0$, $\nabla h_i(\hat{x})d = 0$, $i \in \mathcal{I}$, $\nabla g_j(\hat{x})d = -1$ for some $j \in \mathcal{I}_a(\hat{x})$, and $g_i(\hat{x})d = 0$ $i \in \mathcal{I}_a(\hat{x})$, $i \neq j$. Such a d exist because of the full row rank assumption. Then, Lemma C.1 shows the existence of an $\epsilon > 0$ and function $x(t)$ such that $x(0) = \hat{x}$, $\dot{x}(0) = d$, $h_i(x(t)) = 0$, $i \in \mathcal{I}$, $g_j(x(t)) = -t$ and $g_i(x(t)) = 0$, $i \in \mathcal{I}_a(\hat{x})$, $i \neq j$, for $t \in [-\epsilon, \epsilon]$. In addition, as $x(t)$ is feasible for $t \in [0, \epsilon]$, we must have

$$0 \leq \frac{d}{dt} f(x(t))|_{t=0} = \nabla f(\hat{x})d = -\sum_{i \in \mathcal{E}} \hat{u}_i \nabla h_i(\hat{x})d - \sum_{i \in \mathcal{I}_a(\hat{x})} \hat{y}_i \nabla g_i(\hat{x})d = \hat{y}_j,$$

so that $\hat{y}_j \geq 0$. But this holds for any $j \in \mathcal{I}_a(\hat{x})$. ∎

5. *If $\hat{x}$ is a feasible point such that $\nabla h_i(\hat{x})$, $i \in \mathcal{E}$, are linearly independent, and there exists a d such that $\nabla h_i(\hat{x})d = 0$, $i \in \mathcal{E}$ and $\nabla g_i(\hat{x})d < 0$, $i \in \mathcal{I}_a(\hat{x})$, then $\hat{x}$ is a regular point.*

Proof. Assume that such a d exists, and that the conditions of Theorem B.1 do not hold. We then make use of Corollary B.2, i.e., we show that if there is a $\hat{d}$ that satisfies

$$\nabla f(\hat{x})\hat{d} < 0, \quad \nabla h_i(\hat{x})^T \hat{d} = 0, i \in \mathcal{E}, \quad \nabla g_i(\hat{x})^T \hat{d} \leq 0, i \in \mathcal{I}_a(\hat{x}),$$

then $\hat{x}$ is not a local minimizer. Therefore, the existence of d ensures that $\hat{x}$ is a regular point.

Let $\tilde{d} = a_1\hat{d} + a_2 d$, where a_1 and a_2 are positive constants. We will now show that a_1 and a_2 can be chosen such that $\tilde{d}$ contradicts $\hat{x}$ being a local minimizer.

We have

$$\nabla h_i(\hat{x})\tilde{d} = a_1 \nabla h_i(\hat{x})\hat{d} + a_2 \nabla h_i(\hat{x})d = 0, \quad i \in \mathcal{E},$$

as $\nabla h_i(\hat{x})\hat{d} = 0$ and $\nabla h_i(\hat{x})d = 0$. In addition,

$$\nabla g_i(\hat{x})\tilde{d} = a_1 \nabla g_i(\hat{x})\hat{d} + a_2 \nabla g_i(\hat{x})d < 0, \quad i \in \mathcal{I}_a(\hat{x}),$$

as $\nabla g_i(\hat{x})\hat{d} \leq 0$, $\nabla g_i(\hat{x})d < 0$, $a_1 > 0$ and $a_2 > 0$. Now let $a_1 = -\nabla f(\hat{x})d$ and let $a_2 = (|\nabla f(\hat{x})\hat{d}| + 1)$. Then, $a_1 > 0$, $a_2 > 0$ and

$$\begin{aligned}\nabla f(\hat{x})\tilde{d} &= a_1 \nabla f(\hat{x})\hat{d} + a_2 \nabla f(\hat{x})d \\ &= -\nabla f(\hat{x})d \nabla f(\hat{x})\hat{d} + (|\nabla f(\hat{x})\hat{d}| + 1)\nabla f(\hat{x})d \leq \nabla f(\hat{x})d < 0.\end{aligned}$$

As $\nabla h(\hat{x})$ has full row rank and $\nabla h(\hat{x})\tilde{d} = 0$, Lemma C.1 shows the existence of an $\epsilon > 0$ and function $x(t)$ such that $x(0) = \hat{x}$, $\dot{x}(0) = \tilde{d}$, and $h(x(t)) = 0$ for $t \in [-\epsilon, \epsilon]$. In addition, we have

$$\begin{aligned}\frac{d}{dt} f(x(t))|_{t=0} &= \nabla f(\hat{x})\dot{x}(0) = \nabla f(\hat{x})\tilde{d} < 0, \quad \text{and} \\ \frac{d}{dt} g_i(x(t))|_{t=0} &= \nabla g_i(\hat{x})\dot{x}(0) = \nabla g_i(\hat{x})\tilde{d} < 0, \quad i \in \mathcal{I}_a(\hat{x}).\end{aligned}$$

Consequently, there is an $\epsilon_0 > 0$ such that $f(x(t)) < f(\hat{x})$ for $t \in (0, \epsilon_0)$, and analogously, for each $i \in \mathcal{I}_a(\hat{x})$, there is an $\epsilon_i > 0$ such that $g_i(x(t)) < 0$ for $t \in (0, \epsilon_i)$. Therefore, for

$$0 < t < \min\{\epsilon, \epsilon_0, \min_{i \in \mathcal{I}_a(\hat{x})} \{\epsilon_i\}\},$$

we have $f(x(t)) < f(\hat{x})$, $h_i(x(t)) = 0$, $i \in \mathcal{E}$, and $g_i(x(t)) < 0$, $i \in \mathcal{I}_a(\hat{x})$, showing that $\hat{x}$ is not a local minimizer.

Therefore, if $\hat{x}$ is a local minimizer, $\hat{d}$ cannot exist, and the conditions of Theorem B.1 must hold. ∎

6. If $h(x) = b - Ax$ and g_i, $i \in \mathcal{I}$, are convex, and in addition, $\{x : b - Ax = 0, g_i(x) < 0, i \in \mathcal{I}\} \neq \emptyset$, then every feasible point is regular.

Proof. Let $\widetilde{x}$ be such that $b - A\widetilde{x} = 0$, $g_i(\widetilde{x}) < 0$, $i \in \mathcal{I}_a(\widetilde{x})$. Such an $\widetilde{x}$ exists by the assumption made. Then, for $i \in \mathcal{I}_a(\hat{x})$,

$$0 > g_i(\widetilde{x}) \geq g_i(\hat{x}) + \nabla g_i(\hat{x})(\widetilde{x} - \hat{x}) = \nabla g_i(\hat{x})(\widetilde{x} - \hat{x}),$$

where convexity of g_i and $g_i(\hat{x}) = 0$ has been used. Therefore, letting $d = \widetilde{x} - \hat{x}$, we have $Ad = 0$ and $\nabla g_i(\hat{x})d < 0$, $i \in \mathcal{I}_a(\hat{x})$. The results of 5 may now be applied. (We may assume that A has full row rank, but it is not necessary for linear equality constraints.) ∎

C. Mathematical result

In order to prove our results, we use a result which is analogous to the result used in the proof of Lemma 19.2 in the textbook, but slightly more general. The main step, in using the implicit function theorem, is analogous.

The difficulty arises, because for nonlinear constraints, we have

$$h(\hat{x} + td) = h(\hat{x}) + t\nabla h(\hat{x})d + o(t),$$

where $o(t)$ is a vector-valued remainder term, such that $\lim_{t\to 0} \|o(t)\|/t = 0$, i.e., $h(\hat{x}) + t\nabla h(\hat{x})d$ is the best linear approximation of $h(\hat{x} + td)$ close to $\hat{x}$. For linear constraints, there is no approximation, $o(t) = 0$, and we may apply the analysis directly, as a linear change in x gives a linear change in h. In the nonlinear case, however, we need to know exactly what is the value of $h(\hat{x}) + t\nabla h(\hat{x})d$. The question is then if we can write

$$x(t) = x + td + o(t), \quad \text{with} \quad \lim_{t\to 0} \frac{\|o(t)\|}{t} = 0, \quad \text{and} \quad h(x(t)) = h(\hat{x}) + t\nabla h(\hat{x})d.$$

so that $x(t) \approx \hat{x} + td$, i.e., $x + td$ is the best linear approximation of $x(t)$ for t close to zero and it holds that $h(x(t)) = 0$.

The lemma below shows that if $\nabla h(\hat{x})$ has full row rank, then this can be done. There is an $\epsilon > 0$ and function $u(t)$ such that

$$x(t) = \hat{x} + td + \nabla h(\hat{x})^T u(t), \quad h(x(t)) = h(\hat{x}) + t\nabla h(\hat{x})d \quad \text{for} \quad t \in [-\epsilon, \epsilon],$$

in addition to $u(0) = 0$ and $\dot{u}(0) = 0$. Therefore, $x(0) = \hat{x}$ and $\dot{x}(0) = d$, i.e., $\hat{x} + td$ is the best linear approximation of $x(t)$ for t close to zero. As we only work with functions and gradients, such a best linear approximation will do for our purposes.

Lemma C.1. *Assume that $h : \mathbb{R}^n \to \mathbb{R}^m$ is continously differentiable. In addition, assume that $\hat{x} \in \mathbb{R}^n$ is such that $\nabla h(\hat{x})$ has full row rank. Then, for a given $d \in \mathbb{R}^n$, there is an $\epsilon > 0$ and a continuously differentiable function $u : [-\epsilon, \epsilon] \to \mathbb{R}$, with $u(0) = 0$ and $\dot{u}(0) = 0$, such that $x(t)$ given by*

$$x(t) = \hat{x} + td + \nabla h(\hat{x})^T u(t)$$

satsfies

$$h(x(t)) = h(\hat{x}) + t\nabla h(\hat{x})d, \quad \textit{for} \quad t \in [-\epsilon, \epsilon],$$

in addition to $x(0) = \hat{x}$ and $\dot{x}(0) = d$.

Proof. Let

$$\tilde{h}(u,t) = h(\hat{x} + td + \nabla h(\hat{x})^T u) - h(\hat{x}) - t\nabla h(\hat{x})d.$$

$t \in [-\epsilon, \epsilon]$.
We have $\tilde{h}(0,0) = h(\hat{x}) - h(\hat{x}) = 0$. In addition,

$$\nabla_u \tilde{h}(u,t) = \nabla h(\hat{x} + td + \nabla h(\hat{x})^T u)\nabla h(\hat{x})^T,$$

so that $\nabla_u \tilde{h}(0,0) = \nabla h(\hat{x})\nabla h(\hat{x})^T$. By assumption, $\nabla h(\hat{x})$ has full row rank. Consequently, $\nabla h(\hat{x})\nabla h(\hat{x})^T$ is positive definite, and therefore nonsingular. Therefore $\tilde{h}(u,t)$ is a continuously differentiable function from $\mathbb{R}^{m+1}$ to $\mathbb{R}^m$ such that $\tilde{h}(0,0) = 0$ and $\nabla_u \tilde{h}(0,0)$ is nonsingular. The implicit function theorem then shows the existence of an $\epsilon > 0$ and a continuously differentiable function $u : [-\epsilon, \epsilon] \to \mathbb{R}$, with $u(0) = 0$ such that $\tilde{h}(u(t), t) = 0$ for $t \in [-\epsilon, \epsilon]$.
It remains to show that $\dot{u}(0) = 0$. As $\tilde{h}(u(t), t) = 0$ for t in an interval around 0, we must have

$$\frac{d}{dt}\tilde{h}(u(t), t)|_{t=0} = 0.$$

Differentiation gives

$$\frac{d}{dt}\tilde{h}(u(t), t) = \nabla h(\hat{x} + td + \nabla h(\hat{x})^T u(t))(d + \nabla h(\hat{x})^T \dot{u}(t)) - \nabla h(\hat{x})d.$$

In particular, for $t = 0$, we have $u(0) = 0$, so that

$$0 = \frac{d}{dt}\tilde{h}(u(t), t)|_{t=0} = \nabla h(\hat{x})(d + \nabla h(\hat{x})^T \dot{u}(0)) - \nabla h(\hat{x})d = \nabla h(\hat{x})\nabla h(\hat{x})^T \dot{u}(0).$$

Again, by the full row rank of $\nabla h(\hat{x})$, it follows that $\nabla h(\hat{x})\nabla h(\hat{x})^T$ is positive definite, so that $\dot{u}(0) = 0$. ∎

D. Specific comments regarding equality constraints and inequality constraints

Specifically, in homework assignment 3, you need to make use of Theorem A.1 for a quadratic program, which has convex objective function and linear constraints. The problem is then convex. Theorem A.1 shows that the conditions of the theorem are necessary for local optimality. As the problem is convex, the conditions are also sufficient for global minimality.
Consider the two following convex quadratic programs:

$$(QP_1) \quad \begin{array}{ll} \underset{x \in \mathbb{R}}{\text{minimize}} & \frac{1}{2}x^2 \\ \text{subject to} & 1 - x = 0, \end{array} \qquad (QP_2) \quad \begin{array}{ll} \underset{x \in \mathbb{R}}{\text{minimize}} & \frac{1}{2}x^2 \\ \text{subject to} & 1 - x \leq 0, \end{array}$$

We see by inspection that $x^* = 1$ is a unique global minimizer to (QP_1) as well as to (QP_2).
Now apply Theorem A.1 to see what points satisfy the optimality conditions for each of (QP_1) and (QP_2). Normally, we would use a solver like `quadprog` to get the result, but for this small problem we can solve by inspection. We get, for the two problems, in order,

$f(x) = \frac{1}{2}x^2,\ h(x) = 1 - x,$
1. $\hat{x} - \hat{u} = 0,$
2. $1 - \hat{x} = 0,$
3. $-,$
4. $-,$
Solution: $\hat{x} = 1,\ \hat{u} = 1,$

$f(x) = \frac{1}{2}x^2,\ g(x) = 1 - x,$
1. $\hat{x} - \hat{y} = 0,$
2. $1 - \hat{x} \leq 0,$
3. $\hat{y} \geq 0,$
4. $\hat{y}(1 - \hat{x}) = 0,$
Solution: $\hat{x} = 1,\ \hat{y} = 1.$

Consequently, if we have solved (QP_1) and found an optimal solution $\hat{x} = 1$, the fact that $\hat{u} = 1 \geq 0$ shows that $\hat{x} = 1$ is also an optimal solution to (QP_2), $\hat{x}$ fulfills the optimality conditions of (QP_2) together with $\hat{y} = \hat{u} = 1$. Conditions 2 and 4 of the optimality conditions of (QP_2) are immediately fulfilled, as the constraint is active. Condition 3 is fulfilled by the sign of $\hat{u}$. There is no need to solve (QP_2), we already know the optimal solution.

We also see that if an equality constraint $h(x) = 0$ is changed to an inequality constraint $g(x) \leq 0$, with $g(x) = h(x)$, then optimality condition 2, concerning feasibility, is changed from $h(\hat{x}) = 0$ to $g(\hat{x}) \leq 0$, and $\hat{u}$ is replaced by $\hat{y}$, with the additional condition 3, imposing a sign restriction, $\hat{y} \geq 0$, and the additional condition 4, imposing complementarity, $\hat{y}(1 - \hat{x}) = 0$. (In the literature, as well as in the statement of the homework assignment, you will also find the theorem with no renaming of $g(x)$ to $h(x)$ and of u to y.) The reversed transformation can be made from an inequality constraint $g(x) \leq 0$ to an equality constraint $h(x) = 0$, with $g(x) = h(x)$.

Finally, if we had obtained $\hat{u} < 0$ in (QP_1), then $\hat{x}$ would have been feasible, but not optimal, to (QP_2). We would then have had to solve (QP_2) to get the optimal solution. This would be the case for example for the problems in which the constraint function $1 - x$ is replaced by $x - 1$. We then get, for the two problems, in order,

$f(x) = \frac{1}{2}x^2,\ h(x) = x - 1,$	$f(x) = \frac{1}{2}x^2,\ g(x) = x - 1,$
1. $\hat{x} + \hat{u} = 0,$	**1**. $\hat{x} + \hat{y} = 0,$
2. $\hat{x} - 1 = 0,$	**2**. $\hat{x} - 1 \leq 0,$
3. $-,$	**3**. $\hat{y} \geq 0,$
4. $-,$	**4**. $\hat{y}(\hat{x} - 1) = 0,$
Solution: $\hat{x} = 1,\ \hat{u} = -1,$	Solution: $\hat{x} = 0,\ \hat{y} = 0.$

Think of the Lagrange multiplier as the size of a force given by the constraint. There has to be balance between the force of the objective function and the forces of the active constraints. The size of the force is zero for an inactive constraint. For an equality constraint, the sign of the multiplier is not relevant for the optimality condition to hold, the constraint must be satisfied exactly. For an inequality constraint, the sign is important. Positive multiplier means force making the constraint satisfied exactly, the objective function would decrease if the constraint was violated. Negative multiplier means that the objective function decreases if the constraint is satisfied strictly, i.e., feasible and inactive, and therefore the point is not optimal.